

Exercise 1: Short questions

- (a) Write a function that prints the elements of a list
- (b) Write a function that prints the elements of a list in reverse
- (c) Write your own implementation of the len function that returns the number of elements in a list.

Exercise 2: Copying lists

- (a) Create a list a with some entries.
- (b) Now set b = a
- (c) Change b[1]
- (d) What happened to a?
- (e) Now set c = a[:]
- (f) Change c[2]
- (g) What happened to a?

Now create a function `set_first_elem_to_zero(l)` that takes a list, sets its first entry to zero, and returns the list. What happens to the original list?

Exercise 3: Primes

- (a) Write a function `is_prime(n)` that returns True only if n is prime.
- (b) Note that apart from 2 and 3, all primes are of the form $6k \pm 1$ (though not all numbers of the form $6k \pm 1$ are prime of course). Update your function to use this.
- (c) Write a function that return a list which holds all primes up to n.
- (d) Write a function that returns a list which hold the first n primes.

Exercise 4: Collatz sequence, part 2

Recall the Collatz sequence problem from Experiment 1. Our goal is to find the number $n < 1;000;000$ that leads to the longest Collatz sequence.

- (a) Write a function that for any n, returns its Collatz sequence as a list
- (b) Write a function that finds the integer x that leads to the longest Collatz sequence with $x < n$.

Exercise 5: Pivots

Write a function that takes a value x and a list ys, and returns a list that contains the value x and all elements of ys such that all values y in ys that are smaller than x come first, then we element x and then the rest of the values in ys

For example, the output of `f(3, [6, 4, 1, 7])` should be `[1, 3, 6, 4, 7]`